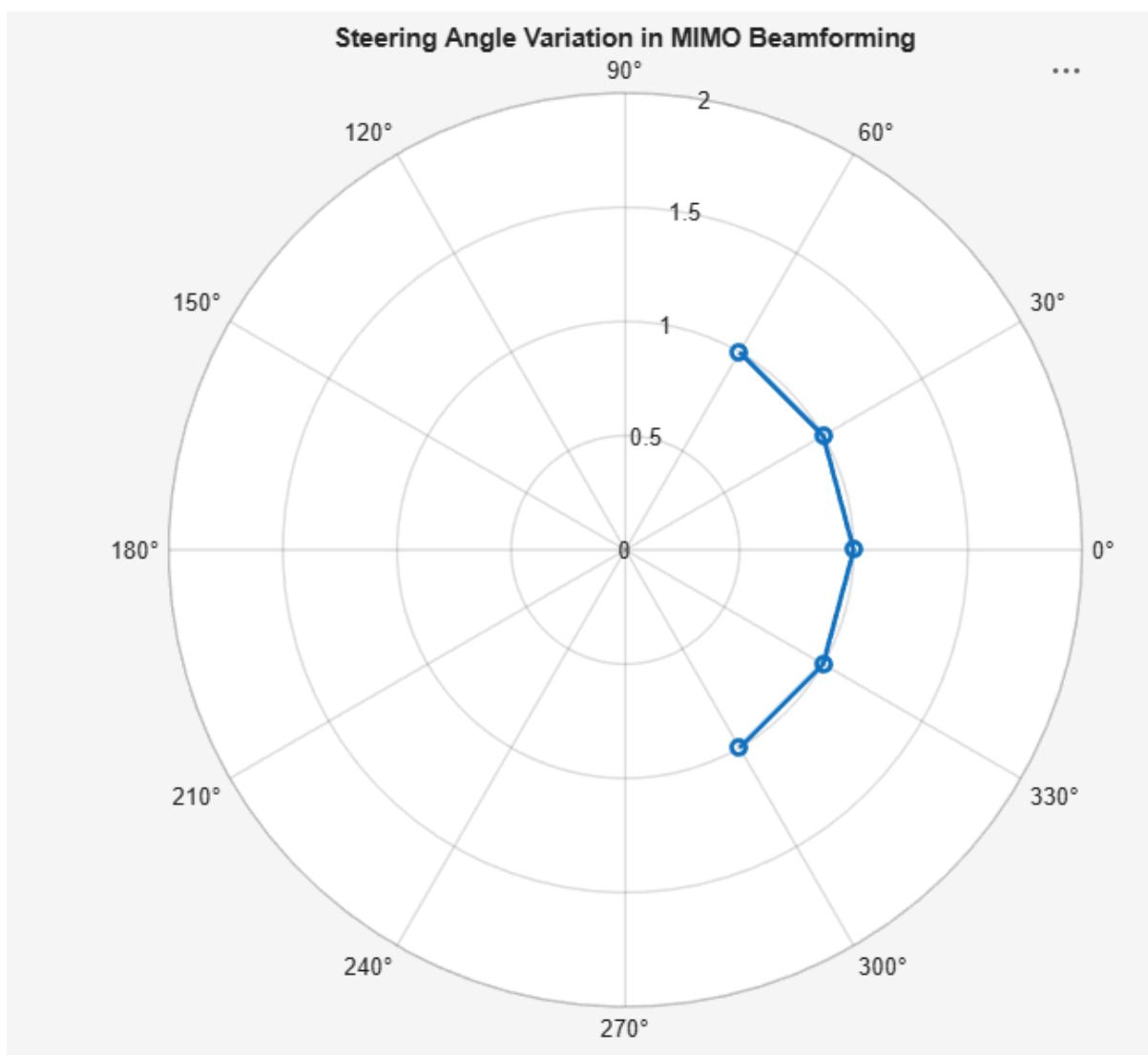


Ex.No.8 Steering Angle Variation in MIMO Beamforming Using MATLAB
Objective 1: Matlab Code and Output

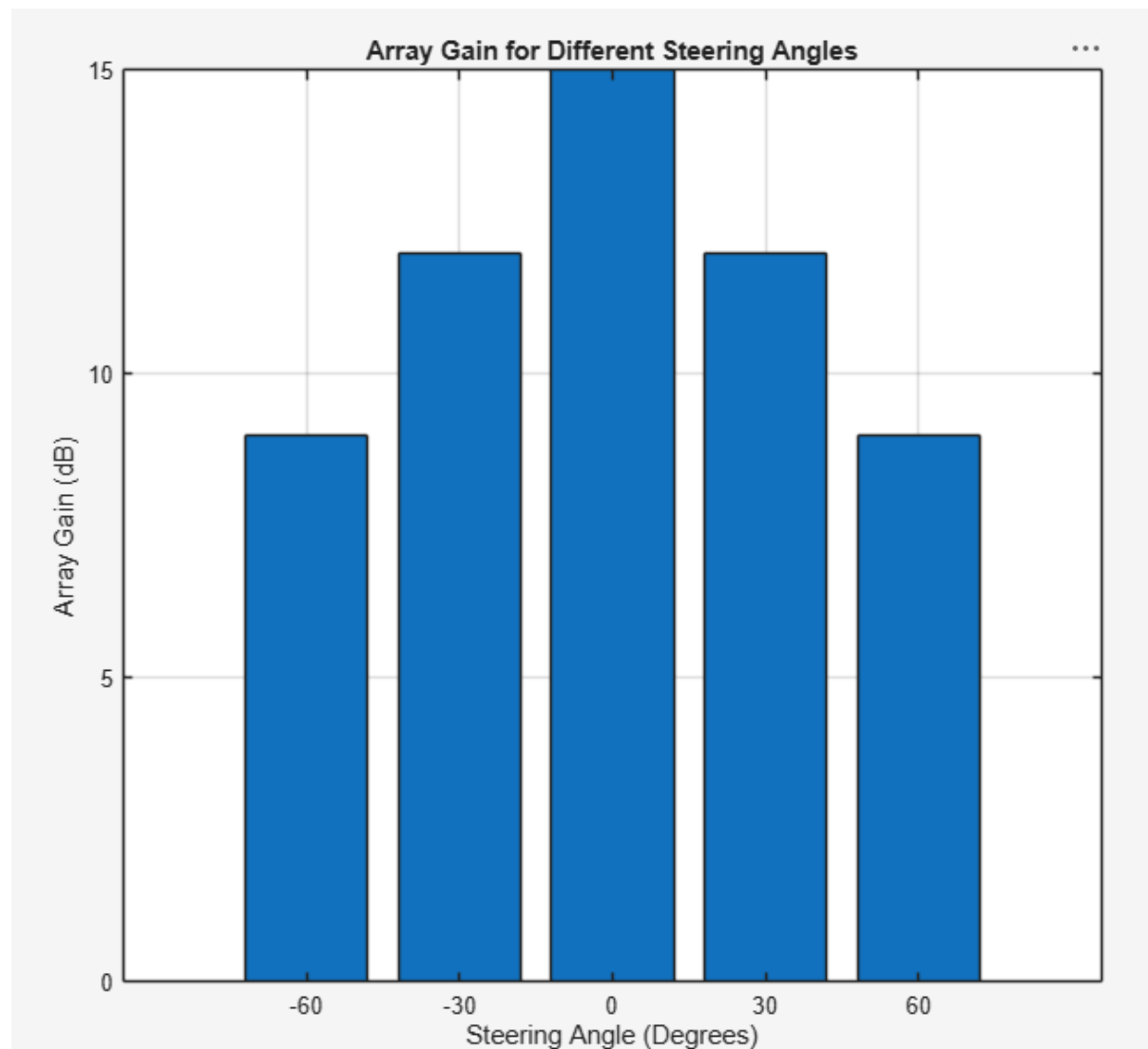
/MATLAB Drive/slotb8.m

```
1   clc;
2   clear;
3   close all;
4   angle = [-60 -30 0 30 60];
5   gain = ones(size(angle));
6   figure
7   % Steering Angles (degrees)
8   % Normalized Gain
9   polarplot(deg2rad(angle), gain, 'o-', 'LineWidth', 2)
10  title('Steering Angle Variation in MIMO Beamforming')
```



Objective 2: Matlab Code and Output

```
12 angle = [-60 -30 0 30 60];  
13 gain = [9 12 15 12 9];  
14 figure  
15 bar(angle, gain)  
16 grid on  
17 % Steering Angles (degrees)  
18 % Array Gain (dB)  
19 xlabel('Steering Angle (Degrees)')  
20 ylabel('Array Gain (dB)')  
21 title('Array Gain for Different Steering Angles')  
22
```



Objective 3: Matlab Code and Output

```
23 desired = [-60 -30 0 30 60];  
24 actual = [-58 -32 1 29 62];  
25 error = abs(desired - actual);  
26 figure  
27 % Desired Steering Angle (degrees)  
28 % Actual Beam Direction (degrees)  
29 % Beam Direction Error  
30 plot(desired, error, 'o-', 'LineWidth', 2)  
31 grid on  
32 xlabel('Desired Steering Angle (Degrees)')  
33 ylabel('Beam Direction Error (Degrees)')  
34 title('Beam Direction Error for Different Steering Angles')
```

